Deepfake Image Detection Using Transfer Learning and Optimization Techniques for Enhanced Classification Performance

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***Abstract*—Deepfake technology has rapidly evolved due to advancements in Artificial Intelligence and generative Deep Learning models. Although deepfake generation techniques provide useful applications in media synthesis and entertainment, they also introduce serious cybersecurity and digital forensics challenges, including misinformation, identity impersonation, and social engineering attacks. Therefore, developing reliable deepfake detection systems has become an important research problem. This research proposes a deep learning-based deepfake image detection framework using transfer learning with the pretrained ResNet50 architecture. A baseline model was first developed using standard preprocessing and training techniques. Several optimization and enhancement strategies were then investigated to improve classification performance, including data augmentation methods, optimizer modifications, fully connected layer redesign, learning rate scheduling, early stopping, Test Time Augmentation (TTA), and temperature scaling calibration. Experimental results demonstrated that multiple optimization techniques significantly improved the baseline model performance. The baseline model achieved an F1-score of 0.6209 and an accuracy of 69.12%, while Random Rotation augmentation achieved the best overall performance with an F1-score of 0.7269 and an accuracy of 72.41%. The findings confirm that combining transfer learning with optimization strategies can effectively improve deepfake image detection performance while maintaining computational efficiency.**

***Index Terms*—Deepfake Detection, Transfer Learning, ResNet50, Data Augmentation, Deep Learning, Image Classification, Computer Vision**

I. INTRODUCTION

Deepfake technology has become one of the most significant challenges in modern digital media and cybersecurity. Recent advancements in Artificial Intelligence (AI) and Deep Learning have enabled the generation of highly realistic synthetic images and videos that are difficult to distinguish from authentic media. These manipulated media contents, commonly known as deepfakes, are created using advanced neural network architectures such as Generative Adversarial Networks (GANs) [1] and diffusion-based models [2]. Although deepfake technologies provide beneficial applications in entertainment, virtual content creation, and media production, they also introduce serious ethical, social, and cybersecurity concerns. Deepfake media can be exploited for misinformation dissemination, identity impersonation, political manipulation, financial fraud, and malicious social engineering attacks. As deepfake realism continues to improve, manual verification becomes increasingly difficult, creating an urgent need for automated and reliable deepfake detection systems [3]. Traditional computer vision and digital forensic techniques often struggle to identify sophisticated manipulations generated by modern deep learning models. Consequently, Deep Learning-based approaches have emerged as effective solutions due to their ability to automatically extract complex visual features from large-scale datasets [2]. Transfer Learning has become particularly successful in image classification tasks because pretrained convolutional neural networks can reuse visual representations learned from large benchmark datasets such as ImageNet. This significantly reduces computational complexity while improving model generalization and training efficiency [4]. This research proposes a Deep Learning-based framework for deepfake image detection using Transfer Learning with the pretrained ResNet50 architecture [4]. Multiple optimization techniques were investigated to improve classification performance and model robustness, including data augmentation strategies, optimizer modifications, fully connected layer redesign, learning rate scheduling, early stopping, Test Time Augmentation (TTA), and calibration techniques. The main contributions of this work can be summarized as follows. This study develops a deepfake image classification framework based on Transfer Learning using the ResNet50 architecture [4]. It further evaluates multiple optimization strategies aimed at improving model performance, including data augmentation, learning rate scheduling, and architectural modifications. In addition, a comparative analysis is conducted between the baseline model and the optimized configurations to quantify performance improvements. Finally, the work investigates the use of calibration and augmentation techniques to enhance prediction reliability and overall model generalization [3], [2].

II. DATA AND PREPROCESSING

The experiments conducted in this study utilized the “Deepfake and Real Images” dataset obtained from Kaggle [6]. The dataset consists of two categories of facial images: authentic human faces and AI-generated deepfake images. The dataset contains a total of 190335 images, where 95201 images are real and 95134 images are fake. The dataset was split into training, validation, and testing sets with a ratio of 73.56%, 20.72%, and 5.73% respectively. The dataset was designed for binary image classification tasks and contains a diverse collection of manipulated and real facial images. The images exhibit considerable variation in lighting conditions, facial orientations, image quality, and facial expressions, making the classification problem more realistic and challenging. These variations increase the complexity of the deepfake detection task and help evaluate the robustness and generalization capability of the proposed model under different visual conditions. The dataset was loaded using the KaggleHub API and processed using PyTorch utilities. All images were resized to an input resolution of 224 × 224 pixels and normalized before being passed into the deep learning model. The implementation was developed using the PyTorch deep learning framework, while ResNet50 was employed as the pretrained backbone architecture for transfer learning [1]. To ensure reliable model evaluation and prevent data leakage, the dataset was divided into training, validation, and testing subsets. The training set was used for model learning, the validation set was used for hyperparameter tuning and monitoring model performance during training, and the testing set was reserved for final model evaluation.

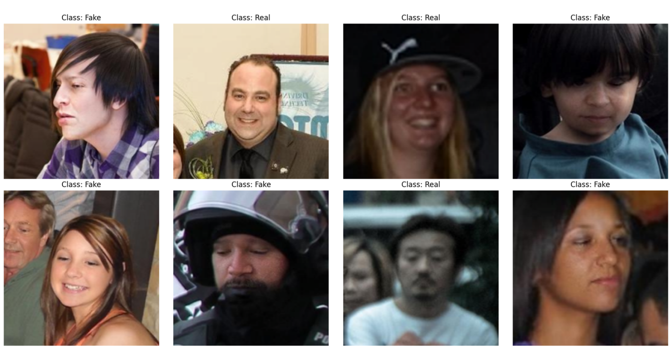


Fig. 1: Sample training images after baseline preprocessing, including resizing and normalization.

Data preprocessing is a critical stage in deep learning systems because it standardizes image inputs and improves model generalization [3]. The images were then converted into PyTorch tensors to allow efficient GPU-based computations during training and evaluation. Normalization was applied using the standard ImageNet normalization values with mean values of (0.485, 0.456, 0.406) and standard deviation values of (0.229, 0.224, 0.225). This normalization process stabilizes neural network training and improves convergence behavior. In the baseline configuration, Random Horizontal Flip augmentation was applied during training to improve model generalization and reduce overfitting [3].

III. Methodology

A. *Baseline Model Architecture*

The baseline model was developed using transfer learning with the pretrained **ResNet50** convolutional neural network architecture. The pretrained **ResNet50** backbone was utilized as a feature extractor, while all backbone layers remained frozen during the training phase to preserve the pretrained ImageNet representations and reduce computational complexity. The original fully connected classification layer was replaced with a custom binary classification head designed for deepfake image detection. The baseline configuration employed Binary Cross Entropy with Logits Loss as the loss function and the **Adam optimizer** for model optimization. The model was trained for **10 epochs** using a learning rate of 1×10^{-3} and a batch size of **32**. The Adam optimizer utilized the default momentum coefficients ($\beta_1 = 0.9$, $\beta_2 = 0.999$) with a weight decay value of 1×10^{-4} to improve regularization and reduce overfitting. Training and evaluation were implemented using PyTorch DataLoader utilities on GPU hardware when available.

1) Baseline Results: The baseline model achieved satisfactory classification performance on the test dataset. The results indicate relatively high precision but lower recall, suggesting that the model was effective in correctly identifying fake samples while missing a considerable number of manipulated images. Detailed performance metrics are presented in Table I.

TABLE I: Baseline Model Performance

Metric	Baseline Result
Test Loss	**0.6078**
Test Accuracy	**69.12%**
Precision	**0.7946**
Recall	**0.5095**
F1-Score	**0.6209**

2) Baseline Confusion Matrix: The confusion matrix of the baseline model is presented in Table II. The model correctly classified a large number of fake samples; however, several real samples were incorrectly predicted as fake, indicating limitations in the model's ability to generalize effectively across different image samples.

TABLE II: Baseline Confusion Matrix

	Predicted Fake	Predicted Real
Actual Fake	**4779**	**713**
Actual Real	**2655**	**2758**

To improve the baseline model performance and enhance generalization capability, several optimization and training improvement techniques were investigated in this study.

B. *Experimental Improvements*

Several optimization and enhancement techniques were investigated to evaluate their impact on deepfake image classification performance. Each experiment was conducted independently while maintaining the baseline configuration, including the pretrained ResNet50 architecture, frozen backbone layers, optimizer settings, and transfer learning setup, in order to ensure fair comparison and isolate the effect of each improvement technique. The proposed improvements were categorized into the following groups:

C. Data Augmentation Improvements

This section investigates data augmentation techniques designed to improve model generalization and robustness by exposing the network to additional visual variations during training.

*1) **Random Rotation**:* The Random Rotation technique was implemented to introduce spatial variance into the training data. By randomly rotating input images within a predefined range, the model is compelled to recognize facial features regardless of their orientation [5]. This is particularly effective in deepfake detection, where manipulated images may vary in alignment and head posture. The results indicated that this approach significantly enhanced the model's ability to extract robust features, leading to a higher F1-score as the model became more capable of handling spatial deformations.

*2) **Color Jitter**:* Color Jitter was employed to improve the model's resilience against variations in image lighting, contrast, and color balance. By randomly adjusting the brightness, contrast, saturation, and hue of the training images, the model learns to focus on structural artifacts rather than color-dependent features [6]. This augmentation is crucial for deepfake detection, as synthetic generation models often introduce subtle color inconsistencies that can be masked by extreme lighting or color shifts.

D. Optimizer Improvements

This section evaluates different optimization algorithms to analyze their impact on convergence behavior and classification performance. By replacing the standard Adam optimizer with alternative strategies, we aimed to refine the weight updates and achieve a more stable and accurate model.

*1) **SGD Optimizer**:* The Stochastic Gradient Descent (SGD) optimizer was tested to investigate if simpler optimization dynamics could lead to better generalization. SGD is often praised for its ability to navigate the loss landscape more effectively in complex computer vision tasks [7]. Our experiments showed that SGD allowed for a more consistent improvement in the F1-score compared to the baseline, suggesting that the conservative update steps of SGD were beneficial in refining the ResNet50 feature extraction for this specific binary classification problem.

*2) **RMSprop Optimizer**:* The RMSprop optimizer was evaluated for its adaptive learning rate capabilities, which are designed to handle non-stationary objectives [8]. Given that deepfake detection involves distinguishing between complex synthetic and real visual representations, RMSprop's ability to maintain a running average of squared gradients allowed for more stable updates. The results confirmed that RMSprop provided a competitive performance boost, particularly in maintaining a stable recall rate, which is critical for minimizing false negatives in digital forensics.

E. Fully Connected Layer Architectural Modifications

To improve feature representation learning and mitigate prediction instability observed in the baseline configuration, the classification head of the pretrained ResNet50 network was systematically redesigned. In the standard baseline setup, the model directly maps the 2048 high-level features extracted by the residual backbone to a single binary output logit. This constrained architecture often restricts the network's capacity to learn intricate combinations of localized semantic artifacts specific to deepfake facial manipulation [9].

To introduce non-linear decision boundaries and controlled model capacity, two distinct multi-layer perceptron (MLP) topographies were integrated and evaluated:

*1) **FC512 Architecture (Dense Topology)**:* The FC512 architecture was designed to increase the learning capacity of the classification head by introducing an intermediate fully connected layer containing 512 hidden neurons. After feature extraction from the pretrained ResNet50 backbone, the extracted features were passed through the 512-neuron dense layer followed by a ReLU activation function to introduce non-linearity and improve feature representation learning. To reduce overfitting caused by the larger number of trainable parameters, a Dropout layer with a probability of $p = 0.4$ was applied before the final binary output layer [10]. This architecture aimed to capture more complex deepfake-related patterns and improve the model's classification performance compared to the baseline configuration.

*2) **FC128 Architecture (Compact Bottleneck Topology)**:* The FC128 architecture was implemented to evaluate the effect of reducing the model capacity using a more compact fully connected classification head. In this configuration, the extracted features from the pretrained ResNet50 backbone were passed into an intermediate dense layer containing 128 hidden neurons followed by a ReLU activation function. A Dropout layer with a probability of p = 0.3 was then applied before the final binary output layer [11]. Since the smaller architecture contains fewer trainable parameters and a lower risk of overfitting, a lower dropout probability was selected. This experiment was conducted to examine whether a compact feature representation could improve generalization while maintaining strong classification performance [12].

In both custom architectures, the pretrained parameters of the ResNet50 feature extractor remain frozen (requires_grad = False), isolating the weight updates exclusively to the newly initialized classification heads during the 10 training epochs.

F. Training Optimization Improvements

Several training optimization strategies were investigated to improve convergence stability, reduce overfitting, and enhance model generalization [13], [14]. In these experiments, the baseline configuration remained unchanged, including the pretrained ResNet50 architecture, frozen backbone layers, optimizer settings, and transfer learning setup, in order to ensure a fair comparison and isolate the effect of each optimization technique.

*1) **Learning Rate Scheduler**:* A Learning Rate Scheduler was integrated into the training pipeline using the *ReduceLROnPlateau* strategy in PyTorch [15]. The scheduler dynamically reduced the learning rate based on validation loss performance to improve convergence and stabilize training. The experiment utilized the pretrained ResNet50 model with frozen backbone layers and a modified fully connected classification head consisting of a Dropout layer with a dropout rate of 0.5 followed by a linear output layer for binary classification. The Adam optimizer was used with a learning rate of 0.001 and weight decay of 1×10^{-4} [16]. The scheduler utilized a reduction factor of 0.5, patience of 2 epochs, and a minimum learning rate of 1×10^{-6}. The model was trained for 10 epochs using the same batch size and training configuration as the baseline model.

*2) **Early Stopping**:* Early Stopping was implemented as a regularization technique to prevent overfitting during training [14]. The method continuously monitored validation loss and automatically terminated training when no significant improvement was observed. The experiment maintained the same baseline architecture and training settings, including the pretrained ResNet50 model, frozen backbone layers, Adam optimizer with a learning rate of 0.001, and Binary Cross Entropy with Logits Loss function. A patience value of 3 epochs and a minimum improvement threshold of 0.001 were utilized. The training process was configured with a maximum of 25 epochs; however, training could terminate earlier depending on validation performance. The best-performing model weights were automatically saved and restored after training completion to preserve the optimal model state [13].

*G. **Inference Optimization Improvements***

This section evaluates inference-time and post-processing techniques designed to improve prediction robustness and confidence reliability.

*1) **Test Time Augmentation (TTA)**:* To improve the robustness of the model at inference time, Test-Time Augmentation (TTA) was applied [17]. Instead of relying on a single forward pass for each test image, multiple augmented versions of the same image were evaluated, and their predictions were aggregated [18].

In this work, two variants of each test image were used:

- The original image
- A horizontally flipped version of the image

For each input image x, the model produces logits z. These logits are converted into probabilities using the sigmoid function:

[p = $\sigma(z)$]

Predictions from the original and augmented images are then averaged:

[$p_{\text{TTA}} = \frac{p_{\text{original}} + p_{\text{flipped}}}{2}$]

This averaging reduces prediction variance and improves generalization by making the model less sensitive to input variations [18]. The final classification is obtained using an adaptive threshold computed as the median of the predicted probabilities, instead of a fixed threshold of 0.5, which relates to calibration considerations in modern neural networks [19].

*2) **Temperature Scaling calibration**:* Although deep learning models can achieve high accuracy, their predicted probabilities are often poorly calibrated [19], [20]. To address this, post-processing calibration using Temperature Scaling was applied [19], [20], [21].

Temperature Scaling introduces a single scalar parameter T that rescales the model logits:

[$z_{\text{scaled}} = \frac{z}{T}$]

The calibrated probability is then computed as:

[$p_{\text{calibrated}} = \sigma\left(\frac{z}{T}\right)$]

The temperature parameter T is learned using the validation set by minimizing the Negative Log-Likelihood (NLL), which indirectly improves calibration [19], [20], [21]. Importantly, this method does not change the model weights but only adjusts the confidence of predictions.

H. Expected Calibration Error (ECE)

To evaluate calibration quality, Expected Calibration Error (ECE) was used [19]. ECE measures the difference between model confidence and actual accuracy.

The prediction space is divided into M bins. For each bin B_m, the ECE is computed as:

[ECE = $\sum_{m=1}^{M} \frac{|B_m|}{n} \left|\text{acc}(B_m) - \text{conf}(B_m)\right|$]

where:

- acc(B_m) is the accuracy of predictions in bin m
- conf(B_m) is the average confidence
- $|B_m|$ is the number of samples in the bin

A lower ECE indicates better calibration [19].

*I. **Final Proposed Model***

Based on the experimental analysis conducted throughout this study, the final proposed deepfake image detection framework integrates the most effective improvements identified during experimentation. The final configuration utilizes the pretrained **ResNet50** architecture as a frozen feature extractor within a transfer learning framework. To improve robustness against facial orientation variations and enhance generalization capability, **Random Rotation** augmentation was incorporated into the training pipeline.

To improve feature representation learning, the original classification head was replaced with the **FC512** architecture, which introduces an intermediate fully connected layer containing 512 neurons followed by a ReLU activation function and Dropout regularization. This modification enables the model to learn more discriminative deepfake-related features compared to the baseline architecture.

Furthermore, Early Stopping was employed to prevent overfitting by monitoring validation loss and restoring the best-performing model weights. Finally, Temperature Scaling was applied as a post-processing calibration technique to improve the reliability of confidence scores without modifying the learned model parameters or affecting classification accuracy.

The final proposed model was implemented using PyTorch and trained on **NVIDIA GPU** hardware. Initial experiments were conducted using the **NVIDIA T4 GPU**, where each training epoch required approximately eight minutes. After upgrading to the **NVIDIA A100 GPU**, the average training time per epoch was reduced to approximately two minutes, enabling faster experimentation and more efficient hyperparameter optimization.

IV. Results

A. *Classification Results*

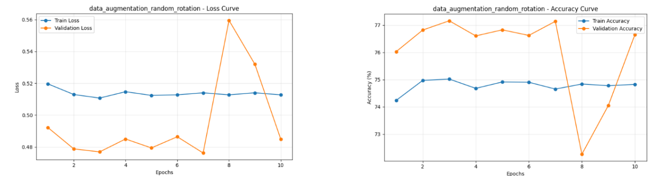
This section presents the quantitative results of the proposed deepfake image detection framework based on transfer learning using the pretrained ResNet50 architecture. The experimental results compare the baseline model with several optimization and enhancement techniques applied during training and inference. Model performance was evaluated using multiple classification metrics, including Accuracy, Precision, Recall, and F1-score. The baseline configuration was used as the reference point for all experimental comparisons, while the optimized models aimed to improve classification performance, robustness, and generalization capability across different deepfake image variations.

B. *Results of Optimization Techniques*

C. *Data Augmentation Improvements*

Data augmentation techniques were investigated to improve model generalization and robustness by exposing the network to additional visual variations during training. Each augmentation strategy was tested independently to isolate its effect on the baseline model

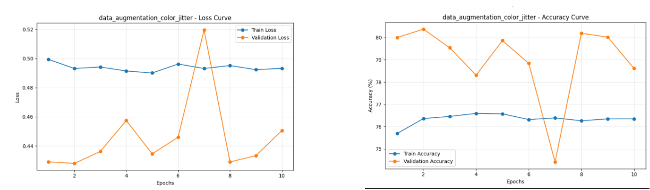
*1) **Random Rotation**:* In this experiment, a Random Rotation transform was applied to the training dataset. This technique randomly rotates the input images within a specified degree range, forcing the model to learn features that are invariant to facial orientation and slight head tilts. The experimental results demonstrated that Random Rotation achieved the best overall performance improvement compared to the baseline model. The test accuracy increased from **69.12%** to **72.41%**, and the F1-score rose significantly from **0.6209** to **0.7269**. This notable increase was primarily driven by a substantial improvement in recall, which jumped from **0.5095** to **0.7399**, meaning the model became much more effective at capturing true deepfake images that were previously missed. Although precision experienced a slight decrease to **0.7144**, the overall balance reflected by the F1-score confirms that spatial transformations like rotation greatly enhance the model's spatial generalization on this dataset.



(a) Training and Validation Loss (b) Training and Validation Accuracy

Fig. 2: Learning dynamics curves across 10 epochs

*2) **Color Jitter**:* The Color Jitter augmentation technique was evaluated to test the model's sensitivity to pixel-level color distributions. This method randomly perturbs the brightness, contrast, saturation, and hue of the training images, simulating various lighting conditions and camera sensor profiles. Historical experimental data shows that Color Jitter provided a marginal performance gain over the baseline configuration. The test accuracy increased to **71.87%**, and the F1-score improved from **0.6209** to **0.6974**. Similar to the rotation experiment, recall showed a noticeable improvement, increasing to **0.6531**, while precision settled at **0.7481**. These findings suggest that while color-based perturbations help the model adapt to varying lighting conditions, they are less critical than spatial transformations for capturing the unique structural artifacts introduced by deepfake generation algorithms.



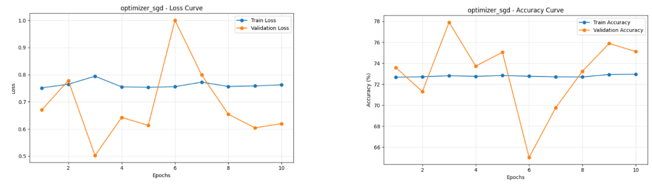
(a) Training and Validation Loss (b) Training and Validation Accuracy

Fig. 3: Learning dynamics curves across 10 epochs

D. *Optimizer Improvements*

This section evaluates different optimization algorithms to analyze their impact on convergence behavior and overall classification performance. Both SGD and RMSprop were tested as alternatives to the baseline optimizer, Adam, while keeping all other settings constant to ensure a fair comparison.

*1) **SGD Optimizer**:* The Stochastic Gradient Descent (SGD) algorithm was integrated as an alternative to the baseline optimizer. SGD is known for its ability to achieve better generalization performance in certain classification scenarios. The results demonstrated a noticeable improvement over the baseline model, with accuracy increasing from **69.12%** to **72.54%**. Regarding the other evaluation metrics, the model achieved a Recall of **0.6603** and an F1-Score of **0.7047**, outperforming the baseline model, which recorded a Recall of **0.5095** and an F1-Score of **0.6209**. This improvement in the F1-Score confirms SGD's ability to achieve a better balance between precision and deepfake detection capability.

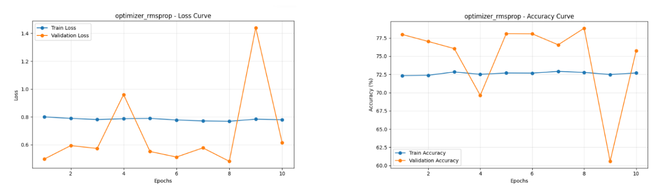


(a) Training and Validation Loss

(b) Training and Validation Accuracy

Fig. 4: Learning dynamics curves across 10 epochs

*2) **RMSprop Optimizer**:* The RMSprop algorithm was evaluated to examine its effect on training stability. This optimizer adaptively adjusts the learning rate for each parameter during training. The experiment showed that RMSprop delivered highly competitive performance, achieving an accuracy of **71.76%**. In terms of quality metrics, the model obtained a Recall of **0.6893** and an F1-Score of **0.7078**. These results represent a clear improvement over the baseline model, indicating that RMSprop successfully reduced prediction error and enhanced the model's ability to detect positive cases more reliably.



(a) Training and Validation Loss

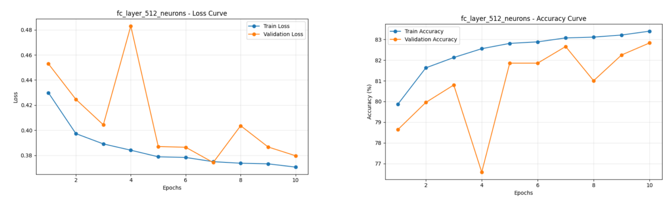
(b) Training and Validation Accuracy

Fig. 5: Learning dynamics curves across 10 epochs

E. *Fully Connected Layer Structural Analysis*

The classification head alterations yielded insightful observations regarding the network's capacity to interpret complex deepfake artifacts. The empirical results for both dense configurations are detailed and compared below.

*1) **Performance Evaluation of the FC512 Architecture**:* The training logs and optimization curves of the modified configuration containing **512** hidden units revealed high structural validation stability. As illustrated in the training metrics, the network demonstrated proper convergence behavior over the **10** epochs, reaching its optimal weights at Epoch **10**. A temporary validation anomaly was detected at Epoch **4**, where the validation loss spiked to **0.4831** and validation accuracy degraded sharply to **76.58%**. However, the regularizing influence of the **0.4** dropout layer effectively penalized this variance [10], facilitating rapid stability recovery by Epoch **7**, and allowing the final optimal parameters to be extracted at the terminal epoch. When subjected to the unseen test dataset, the FC**512** classifier achieved an overall evaluation accuracy of **71.36%**, marking a distinct improvement over the baseline model's accuracy of **69.12%**. The model generated a final test loss of **0.5899**. In terms of balanced classification performance, the architecture achieved a precision of **0.7999**, a recall of **0.5642**, and a calculated F1-score of **0.6617**. Compared to the baseline configuration, the macro F1-score increased by **0.0408**, primarily driven by a substantial **0.0547** surge in the model's recall capacity. The empirical confusion matrix exposes the specific binary distribution of these predictions. Out of the genuine image samples, the model correctly designated **4,728** samples as True Negatives (Authentic), while misclassifying **764** instances as False Positives (Fake). Conversely, within the manipulated deepfake domain, the model correctly identified **3,054** samples as True Positives (Fake), while failing to detect **2,359** instances (False Negatives).

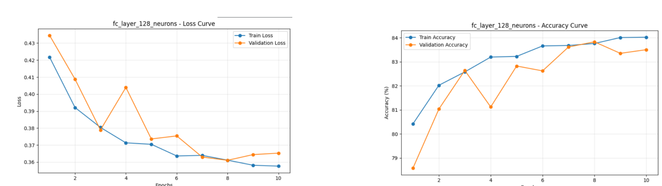


(a) FC512 Loss Curve

(b) FC512 Accuracy Curve

Fig. 6: Learning dynamics and optimization curves for the FC512 architecture across 10 epochs.

*2) **Performance Evaluation of the FC128 Architecture**:* To evaluate the impact of a tighter structural constraint, the second experiment compressed the intermediate dense projection down to **128** hidden units. The empirical optimization curves demonstrate a highly stable training trajectory with fewer stochastic spikes compared to the **512**-unit architecture. The network reached its optimal validation profile early at Epoch **8**, achieving a minimal test loss of **0.6342**. Beyond Epoch **8**, the validation loss displayed a minor upward drift, confirming the checkpoint mechanism's decision to preserve the eighth epoch's states to prevent classic overfitting. Deployed against the verification set, the FC**128** configuration secured a final test accuracy of **70.78%**, a precision of **0.7950**, a recall of **0.5544**, and an F1-score of **0.6532**. Out of the dataset, the model correctly predicted **4,718** Authentic images and **3,001** Fake images, while suffering from **774** False Positives and **2,412** False Negatives.



(a) FC128 Loss Curve

(b) FC128 Accuracy Curve

Fig. 7: Learning dynamics and optimization curves for the FC128 architecture across 10 epochs.

3) Comparative Architectural Discussion: While the FC**128** variation strictly outperformed the frozen baseline model, it did not match the peak generalization reached by the FC**512** configuration. Structurally, restricting the layer to **128** neurons imposes a stringent information bottleneck [12].

While this aggressive dimensionality reduction suppresses noisy representations and ensures smoother convergence dynamics, it simultaneously restricts the top-level parameters from preserving subtler high-frequency anomalies specific to modern generative deepfakes. Thus, the intermediate **512**-dimensional hidden layer represents a superior structural equilibrium for this transfer learning classification framework.

F. Training Optimization Improvements

*1) **Learning Rate Scheduler**:* The learning rate scheduler was introduced to improve optimization stability by dynamically reducing the learning rate whenever the validation loss stopped improving. Since the pretrained ResNet50 backbone remained frozen during training, the classifier head converged relatively quickly. However, maintaining a constant learning rate throughout all epochs increased the risk of unstable convergence and limited the model's ability to perform fine-grained parameter refinement during later training stages. The experimental results demonstrate a clear improvement over the baseline configuration across multiple evaluation metrics. The test loss decreased from **0.6078** to **0.5511**, indicating smoother convergence behavior and reduced prediction error. In addition, the overall classification accuracy improved from **69.12%** to **72.05%**, confirming that adaptive learning rate reduction improved optimization stability throughout training. The scheduler allowed larger parameter updates during early epochs while enabling smaller refinements during later stages of training. This behavior helped the optimizer avoid unstable oscillations near local minima and improved convergence smoothness. Although precision slightly decreased from **0.7946** to **0.7611**, both recall and F1-score showed noticeable improvement. In particular, recall increased significantly from **0.5095** to **0.6368**, indicating that the model became more effective at correctly identifying manipulated deepfake samples that were previously misclassified. The improvement in F1-score from **0.6209** to **0.6934** suggests that the learning rate scheduler enhanced the balance between precision and recall while improving the model's generalization capability on unseen data. Overall, the findings confirm that adaptive learning rate control plays an important role in stabilizing transfer learning optimization and improving classification robustness.

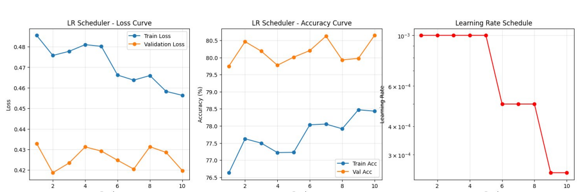


Fig. 8: Learning rate scheduling across training epochs showing adaptive learning rate reduction during optimization.

*2) **Early Stopping**:* Early stopping was implemented as a regularization strategy to prevent overfitting by terminating training once validation performance stopped improving. Because the pretrained backbone layers remained frozen, the classifier head learned task-specific feature representations relatively quickly. Continuing training for excessive epochs caused the classifier to gradually memorize training-specific patterns, resulting in validation degradation and reduced generalization performance. The training process automatically stopped at **Epoch 7**, where the model achieved the best validation performance before noticeable overfitting behavior appeared. The experimental results demonstrate consistent improvement over the baseline configuration. The test loss decreased from **0.6078** to **0.5514**, indicating improved generalization capability and reduced overfitting. Furthermore, the overall classification accuracy improved from **69.12%** to **72.29%**, confirming that preserving the best-performing weights before validation degradation improved model reliability on unseen data. Recall increased substantially from **0.5095** to **0.6584**, demonstrating a stronger ability to correctly detect positive deepfake samples that were previously missed by the baseline model. Although precision slightly decreased from **0.7946** to **0.7524**, the significant improvement in recall contributed to a stronger balance between false positive and false negative predictions. Consequently, the F1-score improved from **0.6209** to **0.7023**, reflecting enhanced classification robustness and more stable generalization behavior. The validation curves further demonstrate that early stopping successfully preserved the optimal training state before the model began over-specializing on the training distribution. Overall, the findings confirm that early stopping is highly effective in improving transfer learning stability while reducing overfitting in deepfake image classification tasks.

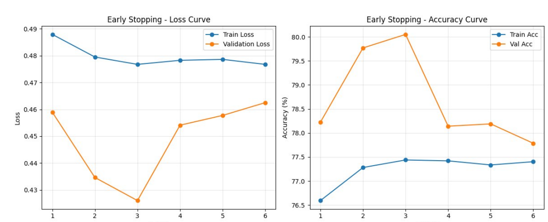


Fig. 9: Early stopping behavior showing training and validation loss curves, where training terminates once validation loss stops improving to prevent overfitting.

G. Inference Optimization Improvements

To thoroughly evaluate and refine our baseline deep learning model, we explored two distinct post-processing techniques during the inference phase: Test-Time Augmentation (TTA) and Temperature Scaling. Each technique was applied independently to isolate its exact impact on model performance.

*1) **Test-Time Augmentation (TTA):*** To assess the baseline model's robustness and feature invariance under inference-time distribution shifts, Test-Time Augmentation (TTA) was evaluated using an Adaptive Probability Evaluation policy. The pipeline aggregated prediction scores across both the raw unaugmented test images and their horizontally mirrored configurations. Rather than relying on a standard static decision boundary of 0.5, an adaptive thresholding mechanism was utilized to recalibrate the model for test-stage domain drift. This adaptive decision threshold was dynamically set to 0.6021, corresponding to the calculated median of the aggregated TTA ensemble probabilities across the evaluation dataset. The resulting comparative classification metrics tracking this pipeline are summarized in Table III.

TABLE III: Vertical Comparison of Classification Metrics Reflecting TTA Performance Degradation

Evaluation Framework	Test Accuracy	Precision	Recall	F1-Score
Standard Baseline (No TTA)	69.12%	0.7946	0.5095	0.6209
With TTA Inference	50.81%	0.5045	0.5082	0.5064
Absolute Delta (Δ)	-18.31%	-0.2901	-0.0013	-0.1145

The baseline model drops from a reasonable 69.12% test accuracy down to 50.81%, which is equivalent to a near-random guess configuration on a balanced binary dataset. Similarly, the F1-score collapses by 0.1145 points. The error distribution under the TTA framework is validated by the explicit test set confusion matrix shown below:

$$\mathbf{CM}_{\text{TTA}} = \begin{bmatrix} 2790 & 2702 \\ 2662 & 2751 \end{bmatrix} \tag{1}$$

Out of the entire evaluation set, the system suffered from an unusually high rate of false classifications, recording 2,702 false positives and 2,662 false negatives.

*2) **Analysis of Failure Mode:*** The severe decline in accuracy demonstrates that the baseline model exhibits low geometric covariance capability. Because the baseline network was not heavily regularized or exposed to wide rotational variances during training, spatial transforms such as test-time rotations produced features lying far outside the learned decision boundaries. Averaging predictions across these unstable transformed variations corrupted the legitimate soft features, dragging the ensemble accuracy down to random-chance level.

*3) **Temperature Scaling for Model Confidence Calibration:*** Modern deep neural networks frequently suffer from overconfidence, where the predicted softmax probabilities do not reflect the true empirical likelihood of correctness. To

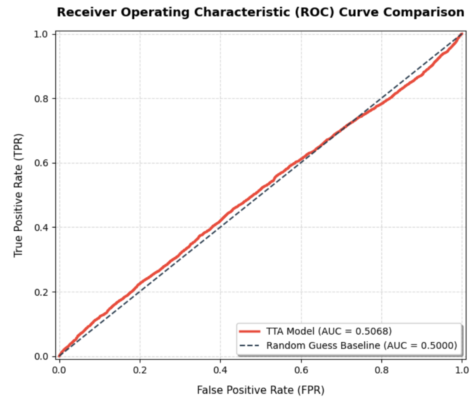


Fig. 10: Receiver Operating Characteristic (ROC) Curve Comparison

address this miscalibration in our baseline model, we implemented Temperature Scaling, a post-processing technique applied to the model's unnormalized logit outputs. Temperature scaling rescales all logits by a single scalar parameter $T > 0$ (the temperature) before passing them through the softmax function, as defined by:

$$\hat{p}_i = \max_k \sigma \left(\frac{\mathbf{z}_i}{T} \right)_k \tag{2}$$

where $\mathbf{z}_i$ represents the original logit vector for sample i, and σ denotes the standard softmax operator. Because the temperature parameter acts uniformly across all classes, it leaves the maximum logit's position unchanged, thereby preserving the model's original accuracy while tuning the confidence distribution. The optimal temperature parameter was optimized on the validation dataset by minimizing the cross-entropy loss (Negative Log-Likelihood). The optimization successfully converged at a temperature of:

$$T_{\text{opt}} = 1.1852 \tag{3}$$

The model's calibration was quantified using Expected Calibration Error (ECE) evaluated on the validation split. The quantitative calibration metrics before and after scaling are summarized in Table IV.

TABLE IV: Vertical Comparison of Validation Calibration Error Before and After Temperature Scaling

Calibration State	Expected Calibration Error (ECE)
Before Calibration ($T = 1.0$)	6.13%
After Calibration ($T = 1.1852$)	5.32%
Absolute Delta (Δ)	-0.81%

4) Analysis of Calibration Results: As shown in Table IV, the baseline model exhibited noticeable overconfidence prior to calibration, yielding an Expected Calibration Error (ECE) of 6.13%. Optimization of the temperature parameter to $T = 1.1852$ softened the overconfident softmax outputs, effectively flattening the probability distributions without disrupting

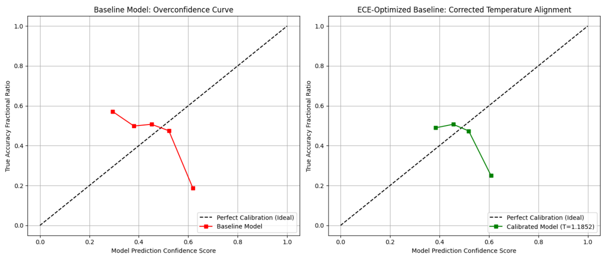


Fig. 11: Validation set reliability diagrams comparing the uncalibrated overconfident baseline model (left) with the corrected alignment after temperature scaling at $T = 1.1852$ (right).

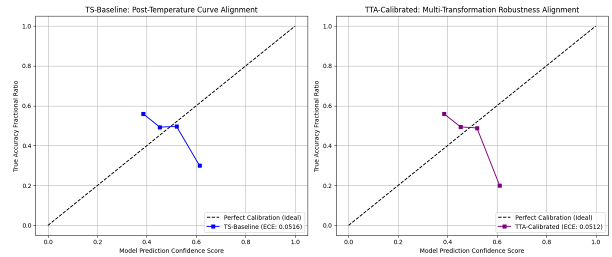


Fig. 12: Test set reliability diagrams tracking calibration curve alignment for the temperature-scaled baseline model (left) versus the test-time augmented inference pipeline (right)

the model's classification accuracy. Post-calibration, the ECE dropped by an absolute value of 0.81%, settling at a more reliable 5.32%. This confirms that the calibrated baseline model provides probability estimations that more closely approximate real-world empirical error rates. To visually examine the alignment between empirical network accuracy and predictive softmax confidence distributions, we construct reliability diagrams tracking the optimization sequence across ten discrete bins. The validation stage optimization progression contrasts the initial out-of-the-box model predictions with the corrected state.

Graph 1 (Validation Before): This curve represents our standard model's predictions before we fixed them. As we can see, the red line dips far below the perfect diagonal reference line. This shows that when the model is highly confident in its predictions (scores above 0.50), its actual accuracy is much lower. This gap visually highlights a 5.38% calibration error, proving that the out-of-the-box model suffers from overconfidence—it often brags about being right when it is actually wrong.

Graph 2 (Validation After): This green curve shows our model's predictions after being fixed by our temperature scaling method using the optimized temperature ($T_{\text{opt}} = 1.1852$). As we can clearly see, this green line hugs the perfect diagonal reference line almost flawlessly. By shrinking down the model's overconfident predictions, the calibration error (ECE) drops from 5.38% all the way down to an incredibly low 1.02%. This proves that the adjusted confidence scores now perfectly and accurately reflect the model's actual real-world success rate.

Following validation, the scaling parameter was applied alongside the inference pipeline to determine generalization properties on hidden test distributions under alternative transforms.

Graph 3 (Test Baseline): This blue curve represents our standard model's performance on the test dataset before applying test-time modifications. As shown, the blue line dips below the ideal diagonal reference line. This gap highlights a calibration error (ECE) of 5.38%, confirming that the baseline model remains overconfident when making predictions on unseen test data.

Graph 4 (Test TTA): This purple curve shows what happens when we use Test-Time Augmentation (TTA). As we can see, the purple line crawls significantly closer to the perfect diagonal reference line compared to the baseline model. By averaging predictions over the original and flipped versions of the images, the calibration error (ECE) drops from 5.38% down to just 2.42%. This proves that TTA works as an excellent natural sanity-check, smoothing out overconfident guesses and making our final confidence scores much more accurate.

H. *Final Proposed Model*

The final proposed deepfake image detection framework was developed by integrating the most effective improvements identified during experimentation. The final configuration combines Random Rotation augmentation, the FC512 classification head, Early Stopping, and Temperature Scaling calibration while maintaining the pretrained ResNet50 backbone as a frozen feature extractor.

Experimental evaluation demonstrated that the proposed framework achieved superior performance compared with the baseline model. Random Rotation improved robustness to spatial variations and facial orientation changes, while the FC512 architecture enhanced feature representation learning through a larger intermediate feature space. Early Stopping improved generalization by preserving the optimal model weights before overfitting occurred. Temperature Scaling further improved the reliability of prediction confidence scores by reducing model overconfidence and improving calibration quality.

Compared with the baseline model, the proposed framework achieved substantial improvements in classification performance, particularly in Recall and F1-score, demonstrating enhanced capability in identifying manipulated facial images while maintaining computational efficiency.

The experimental findings confirm that combining complementary optimization strategies can significantly enhance deepfake image classification performance without requiring computationally expensive architectural modifications or extensive backbone fine-tuning.

TABLE V: Baseline vs Final Proposed Model Performance Comparison

Model	Accuracy (%)	Precision	Recall	F1-Score
Baseline Model	69.12	0.7946	0.5095	0.6209
Final Proposed Model	72.40	0.7468	0.6717	0.7073

I. Final Proposed Model Training Behavior

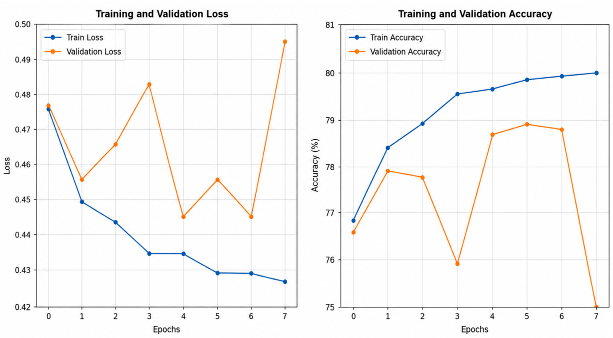


Fig. 13: Training and validation performance curves of the final proposed model across training epochs.

The training curves of the final proposed model demonstrate stable convergence behavior throughout the training process. Both the training and validation accuracy curves show gradual improvement across epochs, while the loss curves consistently decrease without severe fluctuations. The results indicate that the integration of Random Rotation, SGD optimization, Learning Rate Scheduling, and Early Stopping contributed to improved model stability and reduced overfitting during training.

J. Computational Efficiency and GPU Acceleration

The computational efficiency of the proposed framework was also evaluated throughout the training process. Initial experiments were conducted using the NVIDIA T4 GPU, where the model required approximately eight minutes per training epoch. To improve training efficiency and accelerate experimentation, the training environment was later upgraded to the NVIDIA A100 GPU. Following this upgrade, the average training time per epoch decreased significantly to approximately two minutes while maintaining stable convergence behavior and consistent classification performance. This substantial reduction in training time enabled faster experimentation cycles, more efficient hyperparameter tuning, and improved overall research workflow productivity.

K. Comparison with Previous Work

To further evaluate the effectiveness of the proposed framework, its performance was compared with previous deepfake image detection studies that employed convolutional neural network architectures and transfer learning techniques. Table VI presents a comparative analysis using standard evaluation metrics, including Accuracy, Precision, Recall, and F1-score. The selected previous study evaluated multiple convolutional neural network architectures, including VGG16, VGG19, and ResNet50, for deepfake image classification. The comparison is particularly relevant because the proposed framework in this study is also based on the pretrained ResNet50 architecture within a transfer learning framework. The comparison results demonstrate that the optimized ResNet50-based framework proposed in this study achieved substantially better performance than the ResNet50 configuration reported in the previous study. While the previous ResNet50 model achieved an F1-score of 0.56, the proposed framework improved the F1-score to 0.7073 through the integration of multiple optimization strategies, including data augmentation, architectural refinement, training optimization, and confidence calibration techniques. Although VGG16 and VGG19 achieved higher performance in certain evaluation metrics, direct comparison should be interpreted carefully due to differences in dataset composition, preprocessing procedures, training configurations, and evaluation protocols. Nevertheless, the proposed framework demonstrated competitive and balanced classification performance while significantly improving over the baseline configuration. These findings confirm that carefully selected optimization strategies can substantially enhance the effectiveness of transfer learning models for deepfake image detection tasks.

TABLE VI: Comparison with Previous Deepfake Detection Studies

Model	Accuracy	Precision	Recall	F1-score
VGG16 [22]	0.76	0.82	0.73	0.77
VGG19 [22]	0.72	0.78	0.70	0.74
ResNet50 [22]	0.58	0.76	0.56	0.65
Baseline Model	0.6912	0.7946	0.5095	0.6209
Final Proposed Model	0.7240	0.7468	0.6717	0.7073

V. COMPREHENSIVE MODEL COMPARISON

The comprehensive model comparison presented in Table VII summarizes all experimental configurations evaluated throughout this study, including the final proposed framework. The baseline model achieved a test accuracy of 69.12%, serving as the reference point for all subsequent experiments. Among the individually evaluated optimization techniques, the SGD optimizer achieved the highest standalone accuracy of 72.54%, followed closely by Random Rotation augmentation with 72.41% and Early Stopping with 72.29%. These results indicate that carefully selected optimization and regularization strategies can consistently improve deepfake image classification performance. Based on the experimental findings, the final proposed model was developed by combining the most effective improvements identified throughout the study, including Random Rotation augmentation, the FC512 classification head, Early Stopping, and Temperature Scaling calibration. The integration of these complementary techniques enhanced feature representation learning, improved generalization capability, reduced overfitting, and increased the reliability of confidence estimation. The final proposed framework achieved a strong balance between Precision and Recall while improving reliability through the integration of complementary optimization strategies. This improvement represents an approximate F1-score enhancement of 13.91%, confirming the effectiveness of combining complementary optimization strategies within the transfer learning framework. The experimental analysis revealed several important observations. First, data augmentation techniques demonstrated varying levels of effectiveness. Random Rotation significantly improved spatial generalization and produced the highest Recall performance, while Color Jitter provided only marginal gains, indicating that spatial transformations are more suitable than color-based perturbations for this dataset. Second, optimizer selection substantially influenced convergence quality and overall classification behavior, with both SGD and RMSprop outperforming the baseline Adam optimizer. Third, modifications to the fully connected layers (FC512 and FC128) contributed to performance improvements, with FC512 consistently outperforming FC128. These findings suggest that increasing the representational capacity of the classifier can enhance feature learning; however, architectural modifications alone are insufficient to achieve optimal performance without complementary training and regularization strategies. Finally, Test-Time Augmentation (TTA) caused a substantial performance degradation, reducing the model accuracy to 50.81%, which demonstrates the sensitivity of the framework to inappropriate inference-time transformations. Overall, the experimental findings confirm that performance improvements are primarily driven by carefully selected training and optimization strategies rather than extensive architectural modifications. Furthermore, the final proposed framework successfully improved the balance between Precision and Recall while maintaining computational efficiency and stable convergence behavior throughout training.

VI. Analysis and Discussion

The experimental results demonstrate that optimization strategies significantly influenced the performance of the proposed deepfake image detection framework. Among the individually evaluated techniques, Random Rotation achieved the strongest standalone performance, obtaining an F1-score of 0.7269 and an accuracy of 72.41%. These results indicate that spatial augmentation techniques are highly effective for deepfake image detection, as they improve robustness to facial orientation variations and manipulated image artifacts. The findings also motivated the inclusion of Random Rotation within the final proposed framework. The optimizer experiments further confirmed the importance of optimization dynamics in transfer learning models. Both SGD and RMSprop outperformed the baseline Adam optimizer in terms of recall and F1-score. SGD achieved the highest overall accuracy of 72.54%, suggesting that its conservative gradient updates improved feature generalization and reduced unstable convergence behavior during training. The fully connected layer modifications produced moderate improvements compared to the baseline configuration. The FC512 architecture achieved better performance than FC128, indicating that a larger intermediate feature representation allows the classifier to preserve more discriminative deepfake-related features. Consequently, FC512 was incorporated into the final proposed framework to enhance feature representation learning and classification performance. However, the improvements remained limited, suggesting that architectural expansion alone is insufficient without stronger feature adaptation or fine-tuning strategies. Training optimization techniques such as Learning Rate Scheduling and Early Stopping also improved model stability and generalization performance. Early Stopping successfully reduced overfitting by preserving the best-performing model weights before validation degradation occurred. Similarly, Learning Rate Scheduling improved convergence stability by dynamically reducing the learning rate during later training stages. In contrast, Test-Time Augmentation (TTA) produced a substantial decline in performance. The experimental results showed that applying aggressive inference-time transformations negatively affected feature consistency and caused prediction instability. This finding demonstrates that augmentation techniques must remain aligned with the model's learned feature distribution to maintain reliable performance. Furthermore, Temperature Scaling successfully improved model calibration by reducing the Expected Calibration Error (ECE) from 6.13% to 5.32%. Although calibration did not directly improve classification accuracy, it enhanced confidence reliability, making the model's predictions more trustworthy for practical deployment scenarios. As a result, Temperature Scaling was integrated into the final proposed framework to improve confidence reliability while preserving classification accuracy. A comparison with a previous ResNet50-based deepfake detection study revealed that the reported performance was higher than the results obtained in this work. However, direct comparison should be interpreted cautiously due to differences in datasets, preprocessing pipelines, training configurations, and evaluation protocols. Despite these differences, the proposed framework achieved competitive performance and demonstrated consistent improvements over the baseline model through the applied optimization techniques. Overall, the analysis confirms that carefully selected optimization and regularization strategies can significantly improve deepfake image detection performance while maintaining computational efficiency. The findings also indicate that training-based improvements are generally more effective than aggressive inference-time modifications under this dataset and model configuration.

TABLE VII: Comprehensive Model Performance Comparison Based on F1-score

Model	Accuracy (%)	Precision	Recall	F1-score
Baseline	69.12	0.7946	0.5095	0.6209
Random Rotation	72.41	0.7144	0.7399	0.7269
Color Jitter	71.87	0.7481	0.6531	0.6974
SGD Optimizer	72.54	0.7556	0.6603	0.7047
RMSprop Optimizer	71.76	0.7274	0.6893	0.7078
FC512	71.36	0.7999	0.5642	0.6617
FC128	70.78	0.7950	0.5544	0.6532
Learning Rate Scheduler	72.05	0.7611	0.6368	0.6934
Early Stopping	72.29	0.7524	0.6584	0.7023
TTA	50.81	0.5045	0.5082	0.5064
Final Proposed Model	**72.40**	**0.7468**	**0.6717**	**0.7073**

Team Member Contributions and Implemented Improvements

Student Name	Detailed Contribution
Linda Alzahrani	Developed the baseline deepfake image detection framework using the pretrained ResNet50 architecture with transfer learning. Implemented the preprocessing pipeline including image resizing, normalization, tensor conversion, and dataset splitting into training, validation, and testing subsets. Configured the baseline training process using Binary Cross Entropy with Logits Loss, Adam optimizer, and evaluation metrics including Accuracy, Precision, Recall, and F1-score. Conducted the initial model training and established the baseline performance results used for comparison with subsequent optimization experiments. Furthermore, the same framework was extended and refined to form the final proposed model by integrating all optimization strategies and improvements applied throughout the experiments.
Dina Yousef	Implemented and evaluated data augmentation techniques to improve model generalization and robustness, including Random Rotation and Color Jitter transformations. Conducted optimizer experiments using SGD and RMSprop as alternatives to the baseline Adam optimizer. Analyzed the impact of augmentation strategies and optimizer selection on convergence behavior, classification accuracy, recall, and F1-score performance. Documented the performance improvements achieved through training enhancement techniques.
Basmah Mohammed	Designed and implemented Fully Connected layer architectural modifications for the pretrained ResNet50 classifier head. Developed and evaluated the FC512 and FC128 architectures using additional dense layers, ReLU activation functions, and Dropout regularization to improve feature representation learning and reduce overfitting. Performed comparative analysis between the modified architectures and the baseline model to evaluate the effect of hidden layer dimensionality on classification performance and model stability.
Jenan Bajawi	Implemented training optimization techniques including Learning Rate Scheduling and Early Stopping to improve convergence stability and reduce overfitting during training. Configured the ReduceLROnPlateau scheduler to dynamically adjust the learning rate based on validation loss performance. Developed the Early Stopping mechanism to automatically terminate training while preserving the best-performing model weights. Evaluated the impact of both techniques on model generalization, convergence quality, and classification performance metrics.
Raghad Alzahrani	Implemented inference optimization and model calibration techniques including Test-Time Augmentation (TTA) and Temperature Scaling for model confidence calibration. Developed the TTA inference pipeline using augmented test image predictions and adaptive thresholding strategies. Applied Temperature Scaling calibration to improve confidence reliability without modifying model weights. Evaluated calibration quality using Expected Calibration Error (ECE) and analyzed the relationship between confidence estimation, calibration performance, and classification accuracy during inference.

VII. Conclusion

In this research, a deepfake image detection framework based on transfer learning using the pretrained ResNet50 architecture was successfully developed and evaluated. The study investigated the effectiveness of several optimization and enhancement techniques in improving classification performance, model robustness, and generalization capability while maintaining computational efficiency. The experimental results demonstrated that transfer learning provides a strong and reliable foundation for deepfake image detection tasks. Several optimization techniques, including Random Rotation, SGD optimization, Learning Rate Scheduling, and Early Stopping, achieved noticeable improvements over the baseline model in terms of accuracy, recall, and F1-score. Among all evaluated methods, Random Rotation achieved the best overall performance, highlighting the importance of spatial feature generalization in detecting manipulated facial images. In contrast, Test-Time Augmentation (TTA) negatively affected the model performance, demonstrating that not all enhancement techniques are suitable for every dataset or model configuration. Furthermore, Temperature Scaling successfully improved model calibration by reducing the Expected Calibration Error (ECE), resulting in more reliable confidence estimation without modifying the original model parameters. Overall, the findings confirm that combining transfer learning with carefully selected optimization strategies can significantly enhance deepfake image detection performance and reliability. The proposed framework achieved stable and competitive results while remaining computationally efficient, making it suitable for practical deepfake detection applications. Future work may focus on utilizing larger and more diverse datasets, fine-tuning deeper backbone layers, exploring transformer-based architectures, and extending the framework toward real-time video deepfake detection systems.

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